What if you could identify attacks before attackers do?

What is Threat Modeling?

- A structured approach to identify, analyze, and mitigate security threats
- Done during design phase (before coding)
- Focuses on:
- Assets (what to protect)
- Threats (what can go wrong)
- Mitigations (how to prevent it)
- A threat model (diagram + risks + controls)

Importance of Threat Modeling

- Detects vulnerabilities early (low cost fix)
- Helps build secure architecture by design
- Reduces attack surface
- Improves developer + security team collaboration
- Provides clear security requirements for implementation
- Enables risk-based prioritization

Core Concepts

- **Assets:** Sensitive data, APIs, credentials
- **Entry Points:** Login forms, APIs, external integrations
- **Trust Boundaries:** Where data moves between different systems/users
- **Attack Vectors:** Possible paths an attacker can use
- **Mitigations:** Security controls (auth, encryption, validation)

STRIDE Model

- S — Spoofing Impersonating a user/system
- T — Tampering Modifying data or code
- R — Repudiation Denying an action (no logs/evidence)
- I — Information Disclosure Exposing sensitive data
- D — Denial of Service (DoS) Making system unavailable
- E — Elevation of Privilege Gaining higher access than allowed

Types of Threats

- Authentication Attacks → Spoofing
- Data Integrity Attacks → Tampering
- Logging/Tracking Issues → Repudiation
- Data Exposure → Information Disclosure
- Availability Attacks → DoS
- Access Control Issues → Privilege Escalation

AI in Threat Modeling

- **AI helps:**
- Identify hidden attack vectors
- Generate abuse cases automatically
- Analyze large architectures quickly
- **Use cases:**
- Suggest threats from diagrams
- Recommend mitigations
- **Limitation:**
- Needs human validation (false positives possible)

What We Will Do Next?